Customer Segmentation Dashboard

Case Study & Segment Reference Guide

How RFM analysis and KMeans clustering turn raw transactions into targeting decisions

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1. The Business Problem

Most online retailers treat their customer base as one undifferentiated group. Every customer gets the same email, the same discount, the same re-engagement campaign, regardless of whether they spent £20 once or £600,000 across three hundred orders. This wastes marketing budget on customers who were never going to respond, and under-serves the small group of customers actually driving the business.

Knowing average revenue per customer is not enough. An average hides the fact that a small number of customers may be responsible for the majority of revenue, while a large number contribute almost nothing and may already be gone.

This dashboard solves that problem. It scores every customer on Recency, Frequency, and Monetary value (RFM), groups them into five behavioural segments using KMeans clustering, and recommends a specific marketing action for each segment.

2. What the Dashboard Does

The Customer Segmentation Dashboard is a three-tab interactive application built on real UK transaction data from the UCI Online Retail II dataset.

Tab 1 - Segment Overview

A scatter plot of every customer (recency vs monetary value, coloured by segment), a segment profile table, and a revenue-by-segment bar chart. Answers the question 'which segments actually matter?' at a glance.

Tab 2 - Deep Dive

Select a segment to see its top 10 products by revenue, its purchase frequency distribution, and the specific recommended marketing action for that group.

Tab 3 - Model Validation

The elbow plot and silhouette score chart used to justify choosing k=5 clusters, included so the segmentation is not a black box.

3. The Segments Story - What Each Segment Means

Champions

801 customers (15.0% of base) | 68.4% of total revenue

Bought recently, buy often, and spend the most. This small group is responsible for more than two thirds of all revenue in the dataset.

Recommended action: Protect and reward. VIP early access, referral bonuses, exclusive bundles. Any churn here hurts disproportionately.

Loyal

1,530 customers (28.7% of base) | 20.0% of total revenue

High purchase frequency with solid spend, just below Champion level. The largest segment by customer count and the second largest by revenue.

Recommended action: Upsell and nudge upward with bundle deals, milestone rewards, and personalised cross-sell recommendations.

At-Risk

755 customers (14.1% of base) | 6.4% of total revenue

Used to buy regularly and spent well, but have gone quiet recently. A clear win-back opportunity before they are lost for good.

Recommended action: Send a personalised win-back email with a time-limited discount. Act within 30 days.

New Customers

1,239 customers (23.2% of base) | 3.5% of total revenue

Bought recently but have not yet established a habit. Frequency and spend are both still low.

Recommended action: Onboard into a second purchase with a welcome sequence and a second-purchase incentive within 30 days.

Lost

1,011 customers (18.9% of base) | 1.7% of total revenue

Have not purchased in over a year and never spent much when they did. Nearly a fifth of the customer base by headcount, but barely visible in revenue.

Recommended action: One last reactivation email, then retire from active campaigns. Continued outreach wastes budget and hurts deliverability.

4. How the Dashboard Supports Decisions

Where to spend the marketing budget

Champions are 15% of customers and 68.4% of revenue. This single number justifies reallocating retention budget away from blanket campaigns and toward protecting this group specifically.

Where win-back campaigns will actually work

At-Risk customers have a meaningfully higher average spend (£1,243) than New Customers (£410) or Lost customers (£248). Win-back spend is better targeted here than at the Lost segment, which historically never spent much.

Where to stop spending

The Lost segment is 18.9% of the customer base but only 1.7% of revenue. Continued marketing spend on this group has a poor return and risks deliverability problems for the rest of the email program.

Validating the segmentation itself

The Model Validation tab shows a silhouette score of 0.370 at k=5, alongside the elbow plot used to select that value. This lets a stakeholder check the segmentation is statistically defensible rather than taking it on faith.

5. Technical Architecture

Data pipeline

Three pipeline scripts run in sequence:

pipeline/clean.py: loads both sheets of the raw UCI Online Retail II Excel file (~1.06M rows), drops null customer IDs, cancellations, returns, non-UK transactions, and administrative entries (postage, discounts, bank charges).

pipeline/rfm.py: calculates Recency, Frequency, and Monetary value per customer and scores each dimension 1-5 using quintile binning (pandas.qcut).

pipeline/segment.py: log1p-transforms frequency and monetary, scales with StandardScaler, fits KMeans at k=5, and labels each cluster with a business name based on its average RFM scores.

Handling the outlier problem

Raw frequency and monetary values are heavily right-skewed: one customer spent over £600,000 against a median spend in the hundreds of pounds. Clustering on raw values let that single outlier dominate its own cluster, collapsing the Champions segment to one customer. Log1p transformation before scaling fixed this and produced the balanced, business-meaningful segments shown in this report.

Keeping the deployed app lightweight

The dashboard never reads the full 67MB cleaned transaction file. pipeline/segment.py pre-aggregates the top 15 products per segment at build time into a small CSV, which is what the deployed Streamlit app reads. Streamlit Community Cloud only runs dashboard/app.py, never the pipeline, so every file the dashboard depends on is generated ahead of time and committed to the repository.

Stack

Python 3.12 | pandas | scikit-learn | Plotly | Streamlit

6. Segment Results Snapshot

UCI Online Retail II, UK customers, December 2009 to December 2011. 5,336 customers, £14,624,494 total revenue.

Champions - 801 customers (15.0%) - 68.4% of revenue

Avg recency 42 days, avg frequency 23.0 orders, avg spend £12,489

Loyal - 1,530 customers (28.7%) - 20.0% of revenue

Avg recency 64 days, avg frequency 5.9 orders, avg spend £1,911

At-Risk - 755 customers (14.1%) - 6.4% of revenue

Avg recency 397 days, avg frequency 3.5 orders, avg spend £1,243

New Customers - 1,239 customers (23.2%) - 3.5% of revenue

Avg recency 101 days, avg frequency 1.8 orders, avg spend £410

Lost - 1,011 customers (18.9%) - 1.7% of revenue

Avg recency 521 days, avg frequency 1.2 orders, avg spend £248